IR: MCP4725 0x60 -> LM358 -> 2N4401 -> LED

RED: MCP4725 0x61 -> LM358 -> 2N4401 -> LED

J1

MCP4725 IR 0x60

J2

PI I2C INPUT / IR OUT

OUT 1 — DAC_IR — 1 OUT

GND 2 — GND — 2 GND

SCL 3 — SCL — 3 SCL

SDA 4 — SDA — 4 SDA

3V3 5 — 3V3 — 5 3V3

GND 6 — GND — 6 GND

PWR_FLAG

Module ADDR: GND -> 0x60

Both GND pins share 0 V. OUT is a separate signal.

J2: OUT/GND = 2.54 mm; SCL/SDA/3V3/GND = 2.00 mm

J3

MCP4725 RED 0x61

OUT 1 — DAC_RED — 1 J4 1

GND 2 — GND — 2 J4 2

SCL 3 — SCL

SDA 4 — SDA

3V3 5 — 3V3

GND 6 — GND

J4

RED DAC OUT / GND

Module ADDR: VCC -> 0x61

Both GND pins share 0 V. OUT is a separate signal.

5V — J5 1

J5

IR LED A / K

J5 2 — LED_K_IR

5V — J6 1

J6

RED LED A / K

J6 2 — LED_K_RED

R1

10k 1%

1 — 2 CMD_IR

U1A

LM358P / DIP8 socket

3 +

2 −

OUT 1 — AMP_IR

R3

1k

AMP_IR — 1 2 — BASE_IR

Q1

2N4401 / E-B-C

C 3

B 2

E 1

R2

10k 1%

1 — 2 SENSE_IR

R4

100R 1% 0.25W

1 — 2 GND

R6

10k 1%

1 — 2 CMD_RED

U1B

LM358P / DIP8 socket

5 +

6 −

OUT 7 — AMP_RED

R8

1k

AMP_RED — 1 2 — BASE_RED

Q2

2N4401 / E-B-C

C 3

B 2

E 1

R7

10k 1%

1 — 2 SENSE_RED

R9

100R 1% 0.25W

1 — 2 GND

I_LED approximately V_DAC / (2 x 100 ohm)

I_LED approximately V_DAC / (2 x 100 ohm)

C2

DNP loop comp

AMP_IR — 1 2 — SENSE_IR

C2: DNP – DO NOT FIT until loop measured

C4

DNP loop comp

AMP_RED — 1 2 — SENSE_RED

C4: DNP – DO NOT FIT until loop measured

SHARED SUPPLY: U1A + U1B + U1C = ONE LM358 DIP-8

MCP4725 VCC = 3.3 V ONLY; J2 is the only Pi cable input.

ADDR is configured on each module, not by this carrier PCB.

PWR_FLAG — 5V

J7

5V INPUT / GND

1 — 5V

2 — GND

C7

10uF 16V

C5

100nF X7R 50V

U1C

LM358P / DIP8 socket

8 V+

4 V−

PWR_FLAG — GND

H1 M3 / 3.2mm

H2 M3 / 3.2mm

H3 M3 / 3.2mm

H4 M3 / 3.2mm

J2 powers and controls both DAC modules over one shared Pi I2C bus.

OUT pins are DAC outputs: never connect them together.

Matching labels connect SCL, SDA, 3V3 and GND between modules.

For RED: remove the old ADDR-GND bridge before selecting VCC.

C5: ceramic, non-polar. C7: electrolytic, + to 5V.

Nguyen Nhat Huy – Pham Thanh Vy

Sheet: /

File: led_ir_driver.kicad_sch

Title: DATN: PPG–Simulator – LED / IR driver

Size: A3 | Date: | **Rev: 1.8**

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